Module 5 Lab Reflective Journal

Course: ITAI 1371

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This lab helped me better understand why data preparation is such an important part of machine learning. Before completing this lab, I understood that data had to be cleaned but I did not fully understand how missing values, categorical information and different numerical scales could affect a machine learning model. Working through each step helped me see how raw data has to be prepared before it can be used correctly.

One of the first things I learned was how to identify missing values in a dataset. When I examined the Titanic dataset, I saw that the Age column had 177 missing values, the Cabin column had 687 missing values and the Embarked column had 2 missing values. This showed me that datasets can have a large amount of incomplete information. Instead of simply deleting all of those records, I learned how imputation can be used to replace missing values while keeping more of the original data.

For the Age column, I used the median to replace the missing values. I learned that the median can be a better choice that the mean when data may contain outliers or be skewed. After performing the imputation, I checked the Age column again and received a result of zero missing values. Seeing the before and after results helped me understand why it is important to verify that a cleaning method actually worked.

Another concept I learned was One-Hot Encoding. At first, categorical information such as Sex and Embarked is stored as text but many machine learning models need numerical information. Using pd.get_dummies() allowed me to transform those categories into new columns such as Sex_male, Embarked_Q and Embarked_S. This helped me understand how categorical data can be converted into a format that a machine learning model can process.

Feature scaling was another important part of this lab. I used StandardScaler to scale the Age and Fare columns. When I first saw the new negative and decimal values, they looked unusual compared with the original ages and fares. I learned that the values were not incorrect. Instead, StandardScaler transformed them so that the features were measured on a similar scale. This prevents a feature with naturally larger numbers, such as Fare, from appearing more important simply because its values are larger.

Overall, this lab gave me a better understanding of the preprocessing stage of the machine learning workflow. I now understand that preparing data is more than simply loading a dataset. Missing information has to be handled, categorical information may need to be scaled and the results should always be checked before moving forward. I feel more comfortable recognizing these problems in a dataset and choosing an appropriate method to prepare the data for machine learning.

John Grubbs

I enjoyed this assignment because it helped me understand how important data preparation is before building a machine-learning model. Working with the Titanic dataset allowed me to practice handling missing values, converting categorical data into numerical data, and scaling numerical features. These steps showed me that raw data usually needs to be cleaned and transformed before it can be used effectively.

I also gained a better understanding of why different preprocessing techniques are used for different types of data. Imputing missing ages with the median helped preserve the dataset, while one-hot encoding converted the `Sex` and `Embarked` columns into values that machine-learning models can use. Scaling the `Age` and `Fare` columns also showed me how features can be placed on comparable scales. The knowledge check helped reinforce that scaling is generally unnecessary for decision trees because they make splits based on thresholds and the order of values.